I-Logical Approaches
Introduction to Neuro-Symbolic AI

Foundations of Neuro-Symbolic AI

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My background

06/18 - 12/18	**Fritz Haber Institute, Max Planck Society** Data-driven prediction of crystal properties
01/19 - 12/21	**Institute of Mathematics, TU Berlin** PhD in Statistical Learning Theory on: "Uniform Concentration of Tensor and Neural Networks: An approach towards Recovery Guarantees"
since 01/23	**ENEXA (EU Horizon Research Project)** Tensor-network based approaches to reason efficiently on Knowledge Graphs
06/21 - 06/22	**InstruNEXT GmbH** Data-driven optimization of photomask baking processes
07/22 - 12/22	**Schneider Electric SE** Benchmarking of AI-dedicated hardware
01/23 - 06/25	**DATEV eG** Automation of Accounting
since 06/25	**Weierstraß Institute for Applied Analysis and Stochastics** Neuro-Symbolic AI, Quantum Machine Learning

Outline

- Neuro-Symbolic AI
- Use Case: DATEV Automation of Accounting
- Tensors in Artificial Intelligence
- Topics of the course

The future of the software industry

Traditional "symbolic" approach

- Software designed by hard-coded logic
- Requires structured input
- Human-interpretable behavior

AI-based "neural" approach

- Software consists in AI agents based on LLMs
- Flexible input handling
- Black box models

Aim of neuro-symbolic AI

Integrate both approaches:

- Human-interpretable model structure
- Flexibility by controlled LLMs

Strenghts and weaknesses of LLMs

Weaknesses of LLMs

- **Translation:** Translations between natural (e.g. english) and formal (e.g. python) languages
- **Generation:** Creation of new texts similar to existing corpus

Two major concerns about LLMs

Besides having astonishing knowledge representation capabilities, LLMs lack in

- **Explainability:** Problems with hallucinations and formal verifiability
- **Efficiency:** Massive power consumptions during reasoning

Extend the capabilities of LLMs

- State-of-the-art AI is built around the inference of LLMs
- But: Inference of LLMs is not the only thing, in which computers are good at!

Examples:

- Execute an algorithm: Programming languages, e.g. python
- Retrieve knowledge from a structured database: Query languages, e.g. SQL, SPARQL

Aim: Extend the capabilities of LLMs

Integrate inference of LLMs with inference in task-specific formal languages.

The Neuro-Symbolic AI Initiative

Neural AI: Bringing efficiency

- Computations (gradients/inference) organized in layers (sets of neurons with tensor weights)
- Deep layers providing effective representation of data (task-dedicated neurons)

Symbolic AI: Bringing ante-hoc explainability

- Based on human interpretable logic
- Traditional approach to AI

How to bring both paradigms together?

Design neural architectures representing symbolic reasoning.

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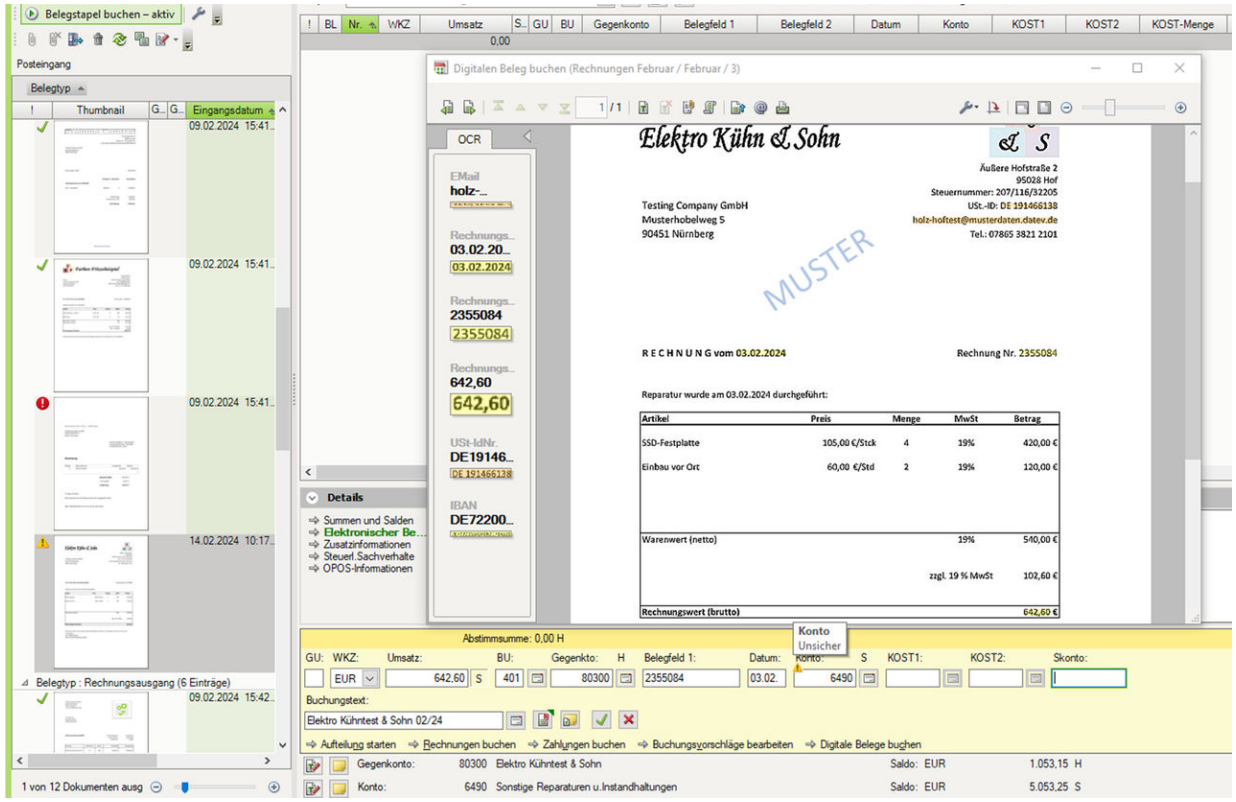
DATEV Automation of Accounting

DATEV, a software company from Nuremberg, sells automation solutions for accounting consultants

- Consultant receives invoices from its customers
- Accounting statements need to be generated

Screenshot from the product[1]:



[1] https://www.datev.de/web/de/steuerberatung/loesungen/rechnungswesen/buchfuehrung-erstellen/automatisierungsservices-rechnungswesen/datev-automatisierungsservice-rechnungen

Requirements

Performance by Neural AI:

- Handle most cases without the need of human supervision
- Understand scenarios based on world knowledge (e.g. "Lindt" in a position indicates choclate consumption)

Trust and manipulations by Symbolic AI:

- Customers need to guarantee certain behaviour, such as respectation of laws
- Manipulation of models based on human input

This is a use case for Neuro-Symbolic AI!

Toy Accounting Example

To showcast the approach we study a toy accounting example. For each invoice three variables are either $\mathtt{True}$ or $\mathtt{False}$:

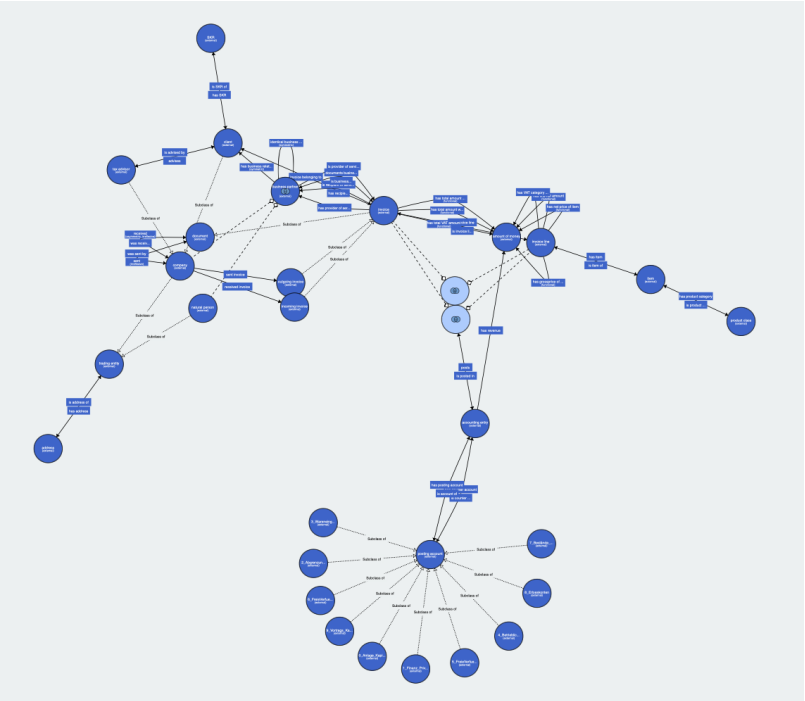
- A Feature is present on the receipt (e.g. whether the receipt was received or sent by the customer).
- The account Account 1 is used to book the receipt.
- The account Account 2 is used to book the receipt.

Let there be two accounting rules to be respected:

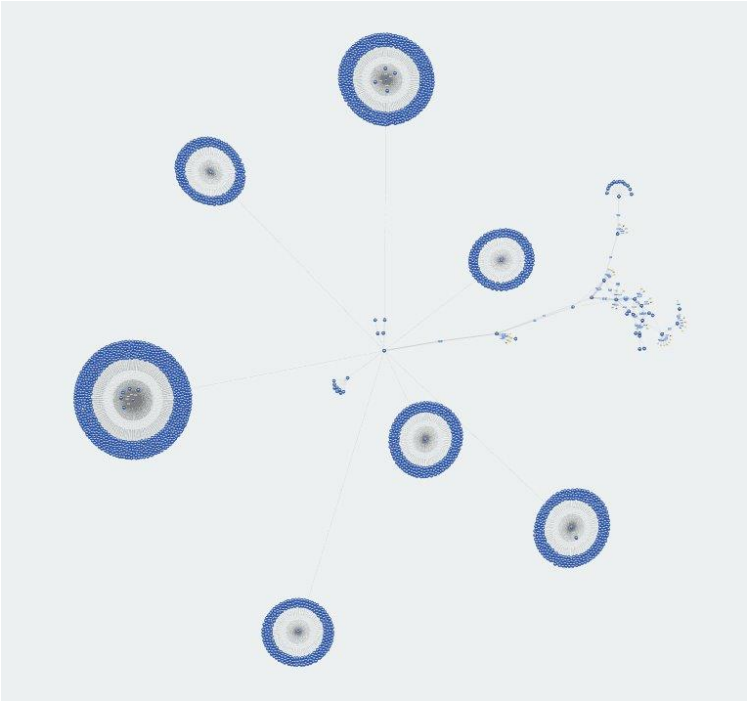
- When the Feature is present on the receipt, the Account 1 needs to be booked.
- Exactly one of the accounts Account 1 and Account 2 is used to book the receipt.

Realistic Problem Sizes

The variables and main relations are visualized in a so-called ontology[2]:



Coarse Visualization



Detailed Visualization

We will come back to ontologies and knowledge graphs later in the course.

[2] https://github.com/EnexaProject/enexa-use-case-1-business-software-solutions

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Tensors: Natural description of factored representations

Factored representations are collection of variables $\{X_0, \ldots, X_{d-1}\}$, each with a finite set of states $\{0, \ldots, m_k - 1\}$. The set of states is then the cartesian product

$$\bigtimes_{k \in \{0,\ldots,d-1\}} \{0, \ldots, m_k - 1\}$$

Knowledge about the system, for example the possibility of each state, is a tensor

$$\tau : \bigtimes_{k \in \{0,\ldots,d-1\}} \{0, \ldots, m_k - 1\} \to \mathbb{R}$$

Factored representation of the toy accounting system

The system has a factored representation by the variables taking values in $\{0, 1\}$ (i.e. $m = 2$):

- X_F: Whether the Feature is present on the receipt.
- X_{A1}: Whether the Account 1 is used.
- X_{A2}: Whether the Account 2 is used.

Recall the two accounting rules:

- When the Feature is present on the receipt, the Account 1 needs to be booked: $X_{A1} \oplus X_{A2}$
- Exactly one of the accounts Account 1 and Account 2 is used to book the receipt: $X_F \Rightarrow X_{A1}$

Coordinatewise tensor visualization

Boolean tensors of leg-dimension $m = 2$ are

$$\tau \; : \; \bigtimes_{k \in \{0,\ldots,d-1\}} \{0,1\} \to \mathbb{R}\,.$$

We visualize such maps as

- $d = 1$: **Vectors**

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- $d = 2$: **Matrices**

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

- $d = 3$: **Order-3 tensors**

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

A computational interface: Tensors in Artificial Intelligence

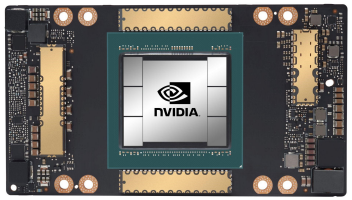
On the software side:

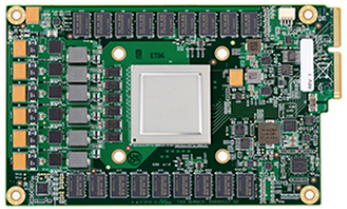
- Parallel computing based on tensor algebra
- Standard ML libraries: TensorFlow, Pytorch

On the hardware side:

- Dedicated for graphic processing: Graphic Processing Units (GPUs)
- AI-dedicated hardware: Tensor Processing Units (TPUs)

TensorFlow





Tensors for neuro-symbolic AI

- **Modelling principle:** Representation of logical and probabilistic models by tensors.
- **Sparsity principle:** Decomposition of tensors into tensor networks.
- **Inference principle:** Contractions of tensor networks.

We orient on the preprint:

A tensor network formalism for neuro-symbolic AI

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Abstract

The unification of neural and symbolic approaches to artificial intelligence remains a central open challenge. In this work, we introduce a tensor network formalism, which captures sparsity principles originating in the different approaches in tensor decompositions. In particular, we describe a basis encoding scheme for functions and model neural decompositions as tensor decompositions. The proposed formalism can be applied to represent logical formulas and probability distributions as structured tensor decompositions. This unified treatment identifies tensor network contractions as a fundamental inference class and formulates efficiently scaling reasoning algorithms, originating from probability theory and propositional logic, as contraction message passing schemes. The framework enables the definition and training of hybrid logical and probabilistic models, which we call Hybrid Logic Networks. The theoretical concepts are accompanied by the python library tnreason, which enables the implementation and practical use of the proposed architectures.

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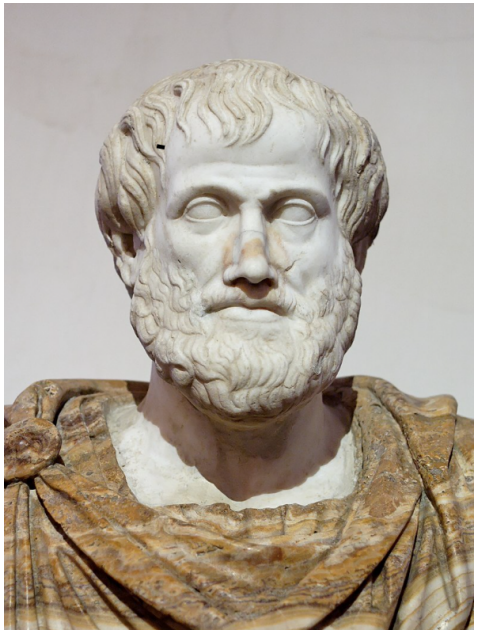
I - Logical Approaches

How to represent knowledge?
$\rightarrow$ **Syntax**

How to infer knowledge from given knowledge?
$\rightarrow$ **Semantics**



Aristotle

Example: When it holds, that

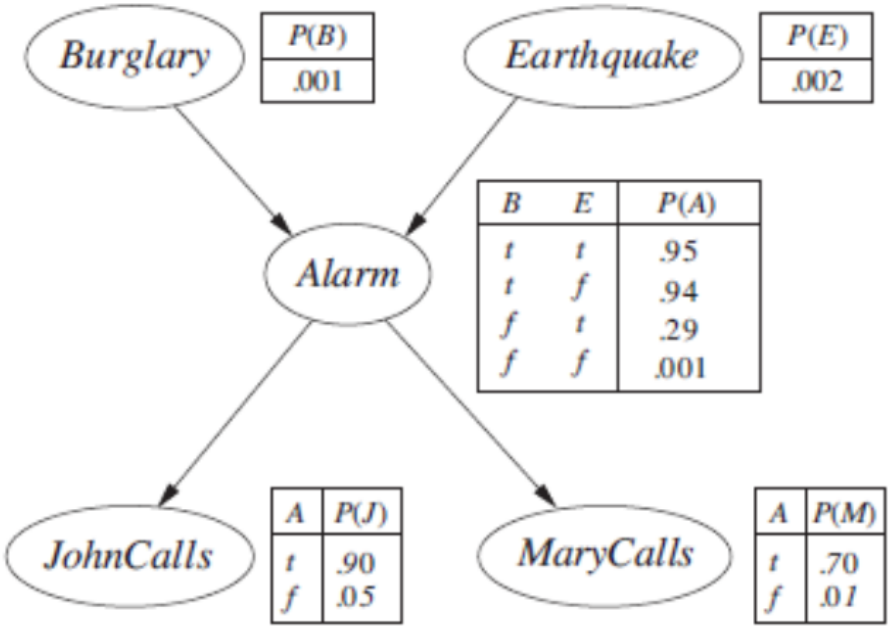
$$\mathrm{hasCampusInSchweinfurt}(\mathrm{THWS}) \quad \text{and}$$
$$\forall x : \mathrm{hasCampusInSchweinfurt}(x) \rightarrow \mathrm{hasCampusInGermany}(x)$$

then also

$$\mathrm{hasCampusInGermany}(\mathrm{THWS})\,.$$

II - Probabilistic Approaches

Bayesian Networks model probabilistic



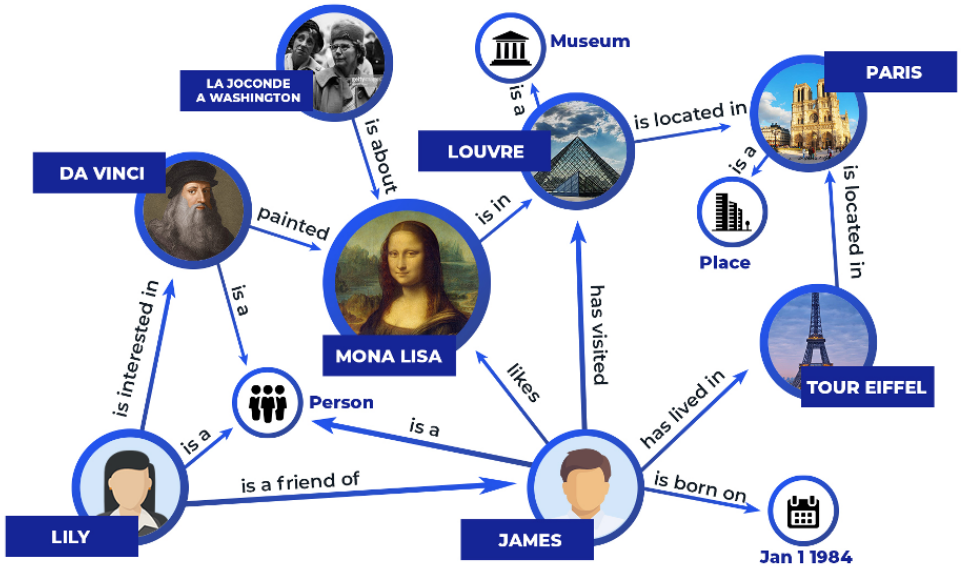
Challenges:

- Learn the dependence structure of the variables
- Extract the causal relations of the variables

III - Statistical Models of Knowledge Graphs

Extension to structured representations:

- Worlds contain objects from classes.
- The objects are enumerated using further tensor axes.



[3]

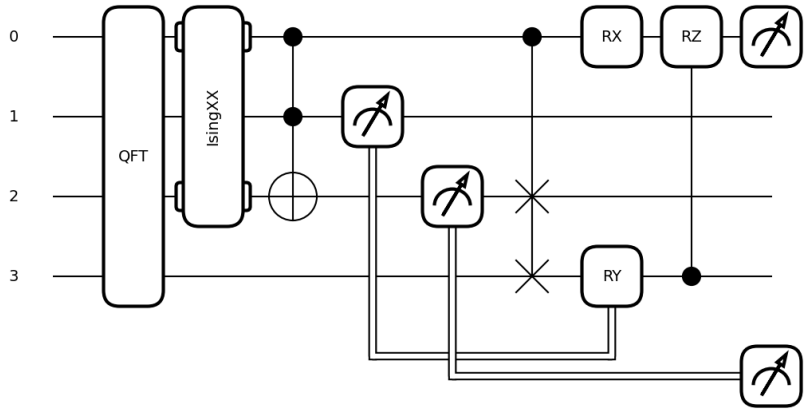
Main goal

Predict the connectivity of objects, using neuro-symbolic statistical models.

[3] https://ml-digest.com/knowledge-graphs/

IV - Quantum Machine Learning

Quantum circuits are tensor networks[4]:



Main goal

Increase the efficiency of reasoning with tensor networks by quantum computation.

[4] https://docs.pennylane.ai/en/stable/introduction/inspecting-circuits.html

Literature

Introductory Textbooks:

- **Russel, Norvig: Artificial Intelligence - A Modern Approach** (Fourth Edition), Pearson Education 2021
- **Koller, Friedman: Probabilistic Graphical Models: Principles and Techniques**, MIT Press 2009
- **Antoniou et al: A Semantic Web Primer** (Third Edition), MIT 2012
- **Schuld, Petruccione: Machine Learning with Quantum Computers**, Springer 2021

Advanced Textbooks:

- **Wainwright, Jordan: Graphical Models, Exponential Families and Variational Inference**, now 2008
- **Hackbusch: Tensor Spaces and Numerical Tensor Calculus**, Springer 2019